

Hypothesis 2: Do BMI Categories Differ in Sleep Duration?

1. Introduction

Body Mass Index (BMI) is widely used to classify individuals into weight categories and is associated with several health outcomes. Sleep duration, an essential factor of wellness, may vary among individuals with different BMI classifications.

This analysis investigates whether **sleep duration differs significantly across BMI categories** in the given dataset.

2. Hypotheses

- **Null Hypothesis (H_0):**
Mean sleep duration is the same across all BMI categories.
 - **Alternative Hypothesis (H_2):**
At least one BMI category has a significantly different mean sleep duration.
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3. Variables Used

- **Dependent Variable:** Sleep Duration (hours)
 - **Independent Variable:** BMI Category
 - Normal
 - Obese
 - Overweight
 - Underweight
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4. Data Cleaning and Preparation

- Removed missing values in *Sleep Duration* or *BMI Category*.
- Converted Sleep Duration into a numeric format.
- Checked for realistic values (1–24 hours).

- Confirmed BMI categories are correctly spelled and consistently labeled.

A subset of the cleaned data:

Sleep Duration BMI Category

6.1	Overweight
6.2	Normal
5.9	Obese
...	...

5. Assumption Testing

5.1 Normality Check (Shapiro–Wilk test)

Results:

- Normal: $p < 0.05 \rightarrow$ **Not normal**
- Overweight: $p < 0.05 \rightarrow$ **Not normal**
- Obese: $p = 0.0571 \rightarrow$ Almost normal
- Underweight: (typically small counts $\rightarrow$ unstable normality)

Since **two groups violate normality**, ANOVA assumptions are not met.

6. Statistical Test Used

Because the normality assumption fails, a **non-parametric** test is required.

👉 Kruskal–Wallis H-Test

(Used to compare the medians of more than two independent groups)

Results:

- H-statistic = 43.0899
- p-value = 2.35×10^{-9}

Interpretation

Since $p < 0.05$, we **reject the null hypothesis**.

✓ Sleep duration differs significantly across BMI categories.

This means individuals in different BMI groups tend to sleep for different amounts of time.

7. Visualizations

7.1 Boxplot

A boxplot was used to compare the distribution of sleep duration across BMI categories. It highlights differences in median values and spread.

8. Conclusion

The analysis provides strong evidence that **BMI classification significantly influences sleep duration**.

This suggests that an individual's body composition may be linked to their sleep behavior and habits.